Spark Gap 2 Build Guide

TLDR;

Dry fit everything before drilling/cutting or soldering anything. When soldering, pots, B7G sockets, and wires that connect to "off board" stuff go *last*.

1. **Ordering:**
    a. Components from the bom.csv in the production folder: Note the footprints to ensure you get the right size (e.g. CP_Radial_D5.0mm_P2.00mm is a 5mm diameter cap with a 2mm pitch between leads).
    b. Tubes (eBay, Tube Depot, Tubes-Store):
        i. 2x EB91/6AL5 tubes These don't need to be "matched"
        ii. 2x B7G PCB mount sockets (7 pin minis)
    c. 1x threaded, DC 5.5mm x 2.1mm jack. Make sure it's one where the case is isolated from the sleeve (e.g. plastic) since 9V pedal power supplies have the + terminal on the sleeve and gnd on the pin. You don't want to short the + to the case.
    d. 2x TR jacks (or TRS). At least one must have a built-in switch, like a Neutrik nmj4hc-s
    e. 2x M3 10mm standoffs
    f. 1x 6 pin footswitch.
    g. A 1590B Hammond enclosure or larger (literally no dimension, h/w/l, should smaller or else things won't fit). If you're not super comfortable with machining the enclosure tightly, a 1590N1 will be more forgiving on placement errors, with roughly the same profile.

2. **Machining enclosure face**. Use the enclosure layout or the C2D file to CNC the top of your enclosure. Or, to manually create (practicing on scrap wood first is a good idea, if you're unsure):
    a. Print out the enclosure layout. Ensure it's to scale by lining up the PCB to the printout
    b. Cut the printout to fit the enclosure, ensure the layout is well centered and flat with no buckling and tape it down. There's very little wiggle room side to side, and not much more top to bottom, so really check this
    c. Use a center punch, or stainless steel nail to tap the center of all the drill holes for which you have bit large enough for
    d. For the remaining holes, using a sharp tool edge, or the tip of a bit, and trace firmly over the outline of the holes
    e. Remove the printout and Drill/cut holes

3. **Machining enclosure sides.** The same enclosure layout has three lines sticking out marking the positions of the holes for the DC power and TR jacks
        i. Mark the enclosure on the face plate at these points
        ii. Draw a vertical line down the sides of the of the enclosure at these points
        iii. Lay your jacks next to these, and mark their centers on the line. You basically want to ensure the jacks have enough room inside the enclosure so
            1. you can screw in the bottom of the enclosure
            2. They don't short out on either the bottom of the enclosure or the top

iv. Before drilling the sides, place the PCB inside the enclosure and line it up, then lay in your jacks inside as well, and make sure they fit where you've marked, then remove everything from the enclosure
v. Use a center punch to mark the hole centers and drill

4. **Assemble step.** Follow this order to minimize issues:
    a. Install all resistors. Most are placed vertically to save space
    b. Install all film caps (i.e. everything but the electrolytics)
    c. Install potentiometers (note that you will have to squeeze between caps in some places)
    d. Install IC, MOSFET, and Diodes
    e. Install the electrolytics
    f. Place the LEDs, but don't solder them yet
    g. Install the 7-pin PCB mount sockets (aka B7G socket)
    h. Push each LED up into the 7-pin socket. Make sure it doesn't protrude from the top of the socket (usually not a problem since the LED's base is wider than the socket's hole). Solder in place.
    i. Jumper the tip to sleeve on the switch side of the input jack
    j. Wire the symmetry switch. Wire should be long to ensure the switch can fit with some space above the "Gain" potentiometer
    k. Wire the jacks to the board
        i. Jacks should face opposite directions
        ii. Make sure there's enough wire so that the jacks can be rotated to fit within the enclosure without shorting to it
    l. Wire the footswitch to the board
        i. Wires should be just long enough that the switch can clear the jacks without tension, again, lay stuff out physically and check
        ii. Pay careful attention to the schematic to avoid mis-wiring it
    m. Strongly suggest you make the DC jack "pluggable" to the PCB, so you can completely disassemble the pedal. If you do, wire up the DC jack now and test. If you don't, then wait till you finish the rest of the assembly.
    n. Loosely attach 10mm standoffs to the PCB
    o. If your TR jacks are isolated (like the nmj4hc-s), solder a wire to the sleeve pin of the input jack, and solder the other end to a washer or locking washer that'll fit over the jack to use as the main ground point.
    p. Dry fit everything into the pedal board. If something doesn't fit, adjust the enclosure accordingly, don't screw anything in yet.
    q. If you didn't make the DC jack pluggable as suggested above, attach the DC plug, and solder it to the board and test.
    r. Screw everything in. Done!